Time Complexity - Full Solutions (Q1-Q30)

This PDF contains explained solutions and complexity results for all questions.

Q1: Single loop runs n times. Complexity O(n).

Q2: Nested loops each run n times. Total n*n = $O(n^2)$.

Q3: Inner loop runs 0+1+...+(n-1) times = n(n-1)/2. $O(n^2)$.

Q4: i doubles every iteration: 1,2,4,...,n. O(log n).

Q5: i halves every iteration: n,n/2,...,1. O(log n).

Q6: Outer n and inner log n. O(n log n).

Q7: n decreases by 1 until 0. O(n).

Q8: n divided by 2 repeatedly. O(log n).

Q9: Two separate O(n) loops. O(n).

Q10: n + n^2 => $O(n^2)$.

Q11: n * n^2 => $O(n^3)$.

Q12: Multiply by 3 each iteration. O(log n).

Q13: $1^2+2^2+...+n^2$ => $O(n^3)$.

Q14: Recursive call fun(n-1). O(n).

Q15: Recursive call fun(n/2). O(log n).

Q16: Two calls to fun(n-1). O(2■).

Q17: Naive Fibonacci recursion. O(2■).

Q18: Factorial recursion. O(n).

Q19: One recursive call per level. O(n).

Q20: Halving recursion. O(log n).

Q21: T(n)=T(n-1)+1 => O(n).

Q22: T(n)=T(n-1)+n => $O(n^2)$.

Q23: T(n)=T(n/2)+1 => O(log n).

Q24: T(n)=2T(n-1)+1 => O(2■).

Q25: T(n)=T(n/2)+n => O(n).

Q26: Sum of log(i) from 1 to n => O(n log n).

Q27: Outer log n, inner n => O(n log n).

Q28: n+(n-1)+...+1 => $O(n^2)$.

Q29: Outer n, inner log n => O(n log n).

Q30: $T(n)=2T(n/2)+1 \Rightarrow O(n)$.